

RESERVOIR DELIVERABILITY

Reservoir deliverability is defined as the oil or gas production rate achievable from reservoir at a given bottom hole pressure. It is a major factor affecting well deliverability. Reservoir deliverability determines types of completion and artificial lift methods to be used. A thorough knowledge of reservoir deliverability is essential for production engineers.

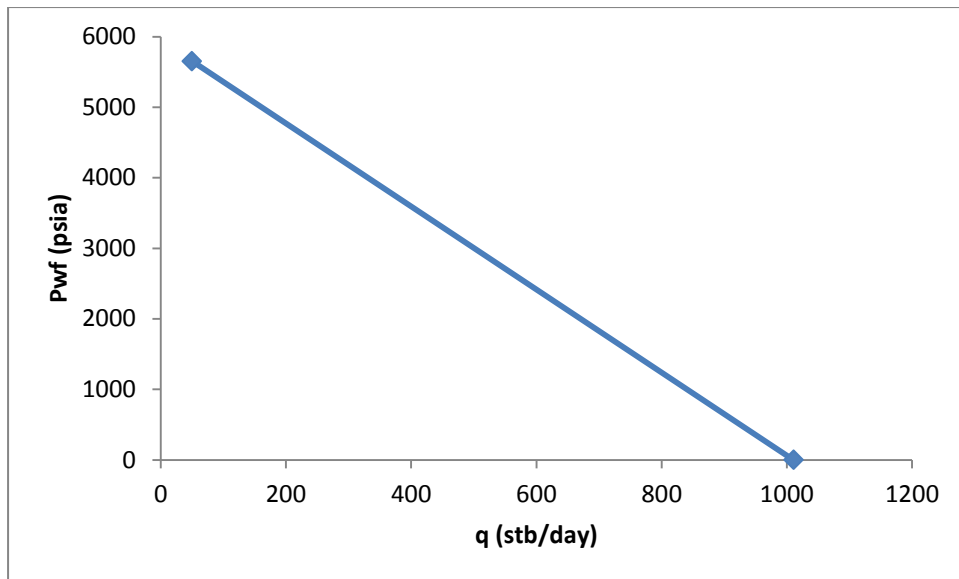
Reservoir deliverability depends on several factors including the following:

- i. Reservoir pressure
- ii. Pay-zone thickness and permeability
- iii. Reservoir boundary type & distance
- iv. Well bore radius
- v. Reservoir fluid properties
- vi. Near well bore conditions
- vii. Reservoir relative permeabilities

Reservoir deliverability can be modelled on the basis of flow regimes such as transient flow, steady state flow and pseudo steady state flow.

INFLOW PERFORMANCE RELATIONSHIP (IPR)

IPR is used for evaluating reservoir deliverability in Production Engineering. The IPR curve is a graphical presentation of the relation between the flowing bottom hole pressure (BHP) and the liquid production rate (q). A typical IPR curve is as shown in the figure below.



The magnitude of the slope of the IPR Curve is called the “productivity index” (P.I or J) that is

$$J = \frac{q}{(P_e - P_{wf})}$$

J is the productivity index, For a single phase flow J is constant, but for two phase flow region, J is not constant.

Well IPR curves are usually constructed using reservoir inflow models, which can be from either theoretical basis or an empirical basis. It is essential to validate these models with test points in the field applications.

IPR FOR SINGLE (LIQUID) –PHASE RESERVOIRS

All reservoir inflow models were derived on the basis of the assumption of a single phase liquid flows. The assumption is valid for under-saturated oil reservoir or reservoir portion where the pressure is above the bubble point pressure.

The following equations define the productivity index (J^*), for flowing BHP above the bubble point pressure as follows:

- i. For radial transient flow around a vertical well.

$$J^* = \frac{q}{p_i - p_{wf}} = \frac{kh}{162.6B_o\mu_o \left(\log t + \log \frac{k}{\phi\mu_o c_t r_w^2} - 3.23 + 0.87S \right)}$$

- ii. For radial pseudo steady state flow around a vertical well

$$J^* = \frac{q}{p_i - p_{wf}} = \frac{kh}{141.2B_o\mu_o \left(\ln \frac{r_e}{r_w} - \frac{3}{4} + S \right)}$$

- iii. For radial pseudo steady state flow around a vertical well

$$J^* = \frac{q}{p_i - p_{wf}} = \frac{kh}{141.2B_o\mu_o \left(\ln \frac{r_e}{r_w} + S \right)}$$

- iv. For pseudo steady state flow around a vertical well

$$J^* = \frac{q}{p_i - p_{wf}} = \frac{kh}{141.2B_o\mu_o \left(\frac{1}{2} \ln \frac{4A}{\gamma C_A r_w^2} + S \right)}$$

Since the productivity index J^* above the bubble point pressure is independent of production rate. The IPR curve for a single (liquid) phase reservoir is simply a straight line drawn from the reservoir pressure to the bubble point pressure.

If the bubble point pressure is 0 psig, the absolute open flow (AOF) is the productivity index J^* multiply by the reservoir pressure.

Question 1.

Construct IPR of a vertical well in oil reservoir, consider

- i. Transient flow at 1 month
- ii. Steady state flow
- iii. Pseudo-steady state flow.

The following data given.

Porosity = $\phi = 0.19$

Effective horizontal permeability = $k = 8.2\text{md}$

Reservoir zone thickness = $h = 53\text{ft}$

Reservoir pressure P_e or $P = 5615\text{psia}$

Bubble point pressure = $P_b = 50\text{ psia}$

Formation formation Volume Factor $B_o = 1.1\text{stb/bbl}$.

Fluid viscosity = $\mu = 1.7\text{cp}$

Total compressibility = $c_t = 0.0000129\text{ psi}^{-1}$

Drainage area = $A = 640\text{ acres}$ ($r_e = 2,980\text{ft}$)

Wellbore radius = $r_w = 0.328\text{ft}$

Skin factor = $S = 0$

SOLUTION

- i. For transient flow, calculated points are:

$$J^* = \frac{q}{p_i - p_{wf}} = \frac{kh}{162.6 B_o \mu_o \left(\log t + \log \frac{k}{\phi \mu_o c_t r_w^2} - 3.23 + 0.87S \right)}$$

$$J^* = \frac{8.2 * 53}{162.6 * 1.1 * 1.7 (\log(30 * 24) + \log \left(\frac{8.2}{0.19 * 0.17 * 0.0000129 * 0.382^2} \right) - 3.23}$$

$$= 0.2075 \text{ stb/d-psi}$$

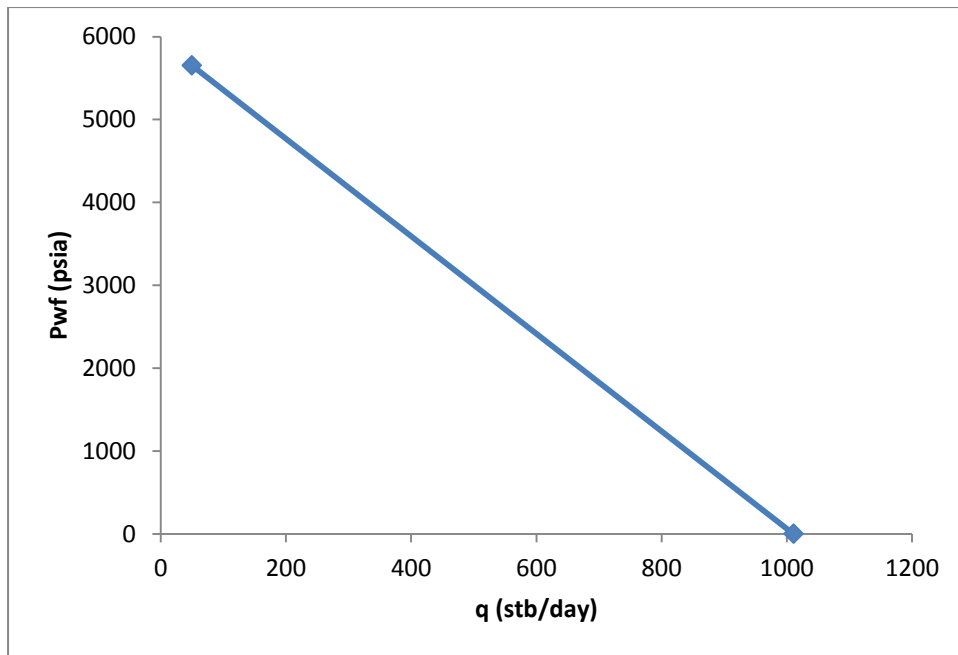
Using the formula

$$J^* = \frac{q}{(P_e - P_{wf})}$$

Calculates the q at p_{wf} of 50 and 5,615 psia (i.e. at bubble points and reservoir pressure), then joins the two points by a straight line. The calculated points are:

Pwf (psia)	q (stb/day)
50	1162
5,651	0

Transient IPR curve



ii. . For pseudo-steady state flow

$$J^* = \frac{q}{P_i - P_{wf}} = \frac{kh}{141.2B_o\mu_o \left(\ln \frac{r_e}{r_w} - \frac{3}{4} + S \right)}$$

$$J^* = \frac{8.2 * 53}{141.2 * 1.1 * 1.7 * \left(\ln \frac{2980}{0.328} - 0.75 \right)}$$

=0.1968 stb/d-psi

Using the formula

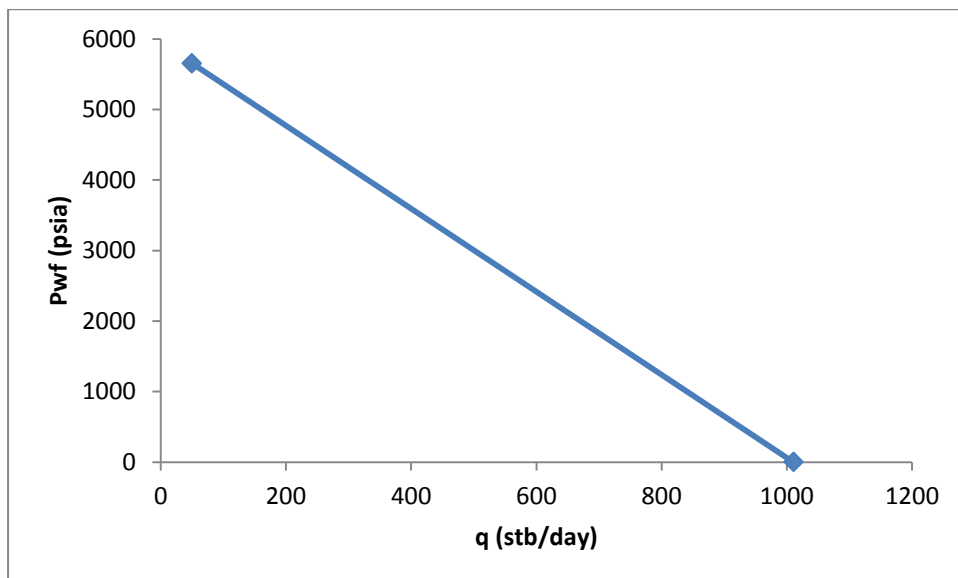
$$J^* = \frac{q}{(P_e - P_{wf})}$$

Calculates the q at p_{wf} of 50 and 5,615 psia (i.e. at bubble points and reservoir pressure), then joins the two points by a straight line.

Calculated points are

P_{wf} (psia)	q (stb/day)
50	1102
5,651	0

Pseudo steady state IPR curve



iii. Steady state IPR curve

$$J^* = \frac{q}{p_i - p_{wf}} = \frac{kh}{141.2 B_o \mu_o \left(\ln \frac{r_e}{r_w} + S \right)}$$

$$J^* = \frac{8.2 * 53}{141.2 * 1.1 * 1.7 * \left(\ln \frac{2980}{0.328} \right)}$$

=0.1806 stb/d-psi

Using the formula

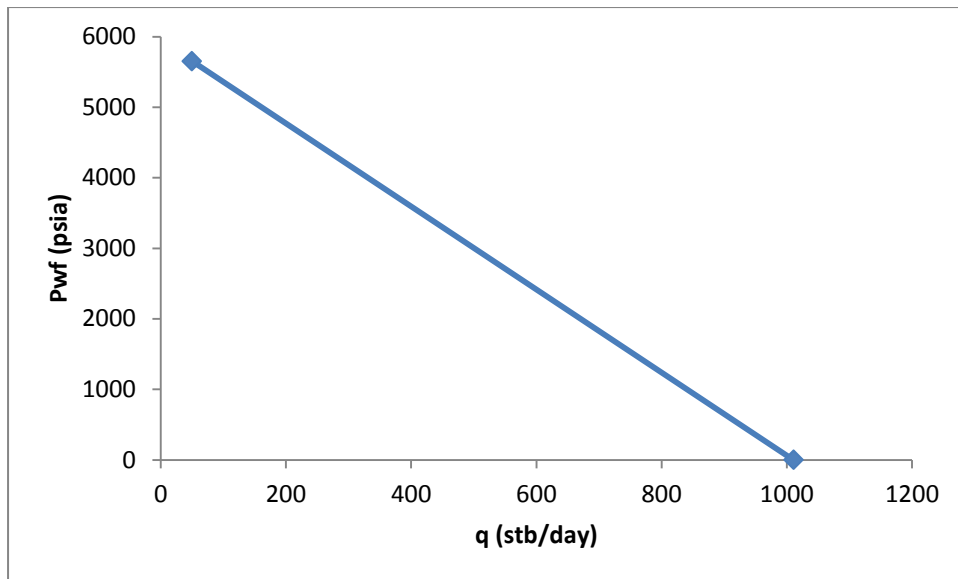
$$J^* = \frac{q}{(P_e - P_{wf})}$$

Calculates the q at p_{wf} of 50 and 5,615 psia (i.e. at bubble points and reservoir pressure), then joins the two points by a straight line.

The calculated points are

Pwf (psia)	q (stb/day)
50	1011
5,651	0

Steady state IPR curve



IPR FOR TWO-PHASE RESERVOIRS

The linear IPR model presented earlier is valid for pressure values as low as bubble point pressure, below the bubble point, the solution gas escapes from the oil and become free gas. The free gas occupies some portion of the pore space which reduces flow of oil. This effect is quantified by the reduced relative permeability. Also oil viscosity increases as its solution gas content drops. The combination of the relative permeability effect and the viscosity effect result in lower oil production rate at a given bottom hole pressure (BHP), the lower the pressure, the larger the deviation. If the reservoir pressure is below the initial bubble point pressure, oil and gas two phase flow exist. In the whole reservoir domain and the reservoir is referred to as a two-phase reservoir.

Only empirical equations are available for modelling IPR of two phase reservoirs. Some of the available empirical equations include:

- a. Vogels (1968) extended by Standings
- b. The Fetkovich equation (1973)
- c. Bandakhlia and Aziz (1989) equation
- d. Zhang (1992) equation

e. Retnanto and Economides (1998) equation

Vogel's equation is still widely used in industry. It is written as:

$$q = q_{\max} \left[1 - 0.2 \left(\frac{P_{wf}}{P} \right) - 0.8 \left(\frac{P_{wf}}{P} \right)^2 \right]$$

Where $q_{\max}$ is an empirical constant. Its value represents the maximum possible value of reservoir deliverability or absolute open flow (AOF). The $q_{\max}$ can be theoretical estimated based on reservoir pressure and productivity index above the bubble point pressure. The pseudo steady state flow follows that

$$q_{\max} = \frac{J * \bar{P}}{1.8}$$

Fetkovich's equation is written as

$$q = q_{\max} \left[1 - \left(\frac{P_{wf}}{\bar{P}} \right)^2 \right]^n$$

or

$$q = c \left(\bar{P}^2 - P_{wf}^2 \right)^n$$

Where c and n are empirical constants and is related to $q_{\max}$ by

$$c = \frac{q_{\max}}{2n}$$

Fetkovich's equation with two constant is ore accurate than Vogel's equation IPR modelling .

Question 2

Construct IPR of a vertical well in a saturated oil reservoir using Vogel equation. The following data are given,

Porosity = $\phi = 0.19$

Effective horizontal permeability = $k = 8.2\text{md}$

Reservoir zone thickness = $h = 53\text{ft}$

Reservoir pressure P_e or $P = 5615\text{ psia}$

Bubble point pressure = $P_b = 5615\text{ psia}$

Formation Volume Factor $B_o = 1.1\text{stb/bbl.}$

Fluid viscosity = $\mu = 1.7\text{cp}$

Total compressibility = $c_t = 0.0000129\text{ psi}^{-1}$

Drainage area = $A = 640\text{ acres}$ ($r_e = 2,980\text{ft}$)

Wellbore radius = $r_w = 0.328\text{ft}$

Skin factor = $S = 0$

Solution

$$J^* = \frac{q}{p_i - p_{wf}} = \frac{kh}{141.2B_o\mu_o \left(\ln \frac{r_e}{r_w} - \frac{3}{4} + S \right)}$$

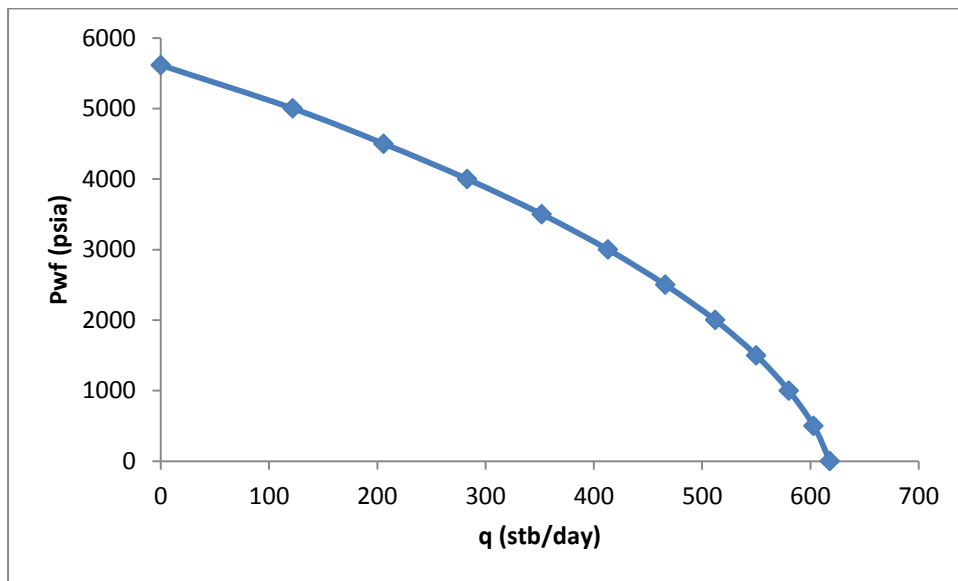
$$= \frac{8.2 * 5.3}{141.2 * 1.1 * 1.7 * \left(\ln \left(\frac{2980}{0.328} \right) - 0.75 \right)}$$

=0.1968 stb/psi.

$$q_{\max} = \frac{J * \bar{p}}{1.8} = \frac{0.1968 * 5615}{1.8} = 618 \text{ stb/day}$$

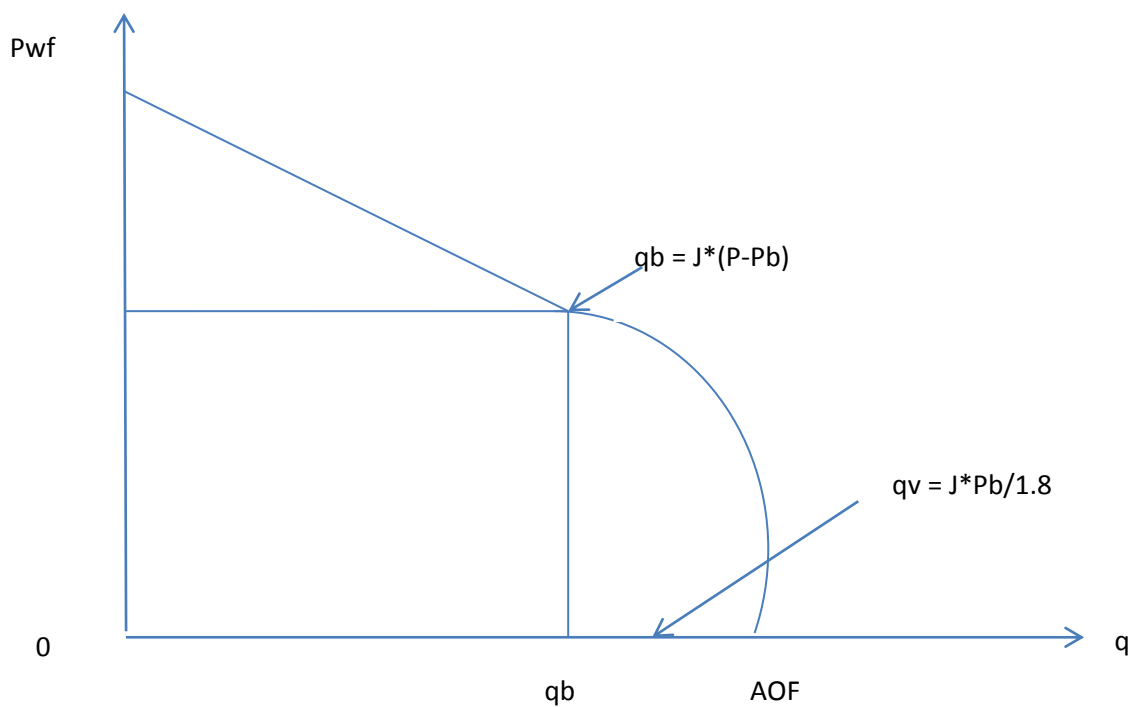
P _{wf} (psi)	q _o (stb/day)
5615	0
5000	122
4500	206
4000	283
3500	352
3000	413
2500	466
2000	512
1500	550
1000	580
500	603
0	618

IPR OF A VERTICAL WELL IN A SATURATED OIL RESERVOIR USING VOGEL EQUATION



IPR FOR PARTIAL TWO PHASE OIL RESERVOIRS

If the reservoir pressure is above the bubble point pressure and the flowing bottom hole pressure is below the bubble point pressure a generalized IPR model can be formulated. This can be done by combining the straight line IPR model for single phase with Vogel's IPR model for two phase flow



According to the linear IPR model, the flow rate at bubble point pressure is

$$q_b = J^* \left(\bar{P} - P_b \right)$$

Based on Vogel IPR model, the additional flow rate caused by a pressure below the bubble point is expressed as

$$\Delta q = q_b + q_v \left[1 - 0.2 \left(\frac{P_{wf}}{P} \right) - 0.8 \left(\frac{P_{wf}}{P} \right)^2 \right]$$

Since

$$q_v = \frac{J^* P_b}{1.8}$$

The equation becomes

$$\Delta q = J^* \left(\bar{P} - P_b \right) + \frac{J^* P_b}{1.8} \left[1 - 0.2 \left(\frac{P_{wf}}{P} \right) - 0.8 \left(\frac{P_{wf}}{P} \right)^2 \right]$$

Question 3

Construct IPR of a vertical well in an under-saturated oil reservoir using the generalized Vogel's equation

The following data are given,

Porosity = $\phi = 0.19$

Effective horizontal permeability = $k = 8.2 \text{ md}$

Reservoir zone thickness = $h = 53 \text{ ft}$

Reservoir pressure P_e or $P = 5615$ psia

Bubble point pressure = $P_b = 3000$ psia

Formation Volume Factor $B_o = 1.1$ stb/bbl.

Fluid viscosity = $\mu = 1.7$ cp

Total compressibility = $c_t = 0.0000129$ psi⁻¹

Drainage area = $A = 640$ acres ($r_e = 2,980$ ft)

Wellbore radius = $r_w = 0.328$ ft

Skin factor = $S = 0$

Solution

$$J^* = \frac{q}{P_i - P_{wf}} = \frac{kh}{141.2 B_o \mu_o \left(\ln \frac{r_e}{r_w} - \frac{3}{4} + S \right)}$$

$$= \frac{8.2 * 5.3}{141.2 * 1.1 * 1.7 * \left(\ln \left(\frac{2980}{0.328} \right) - 0.75 \right)}$$

= 0.1968 stb/psi.

$$q_b = J^* \left(\bar{P} - P_b \right)$$

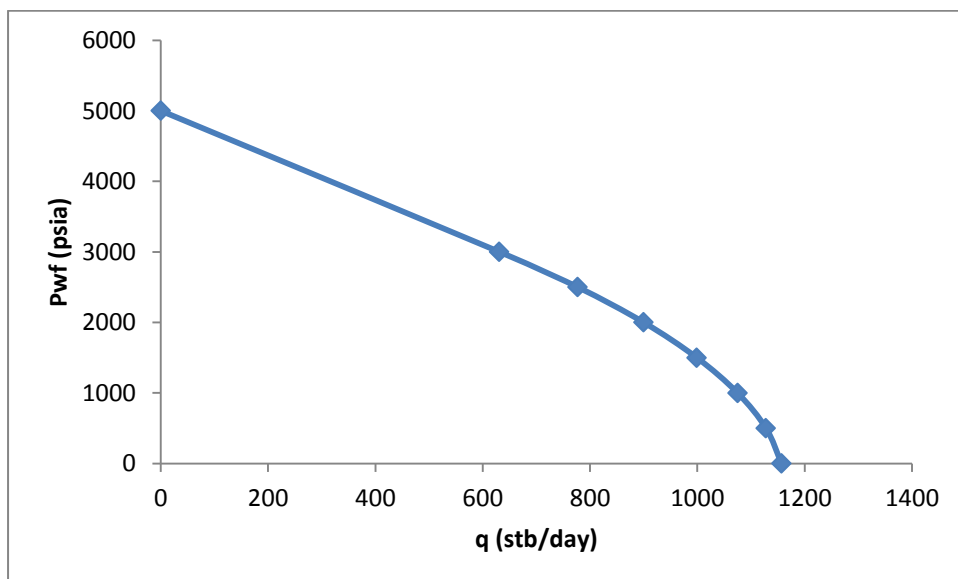
$$= 0.1968 (5615 - 3000)$$

= 5222 stb/day

$$q_v = \frac{J * \bar{p}}{1.8} = \frac{0.1968 * 3000}{1.8} = 328 \text{ stb/day}$$

Calculated points are

P_{wf} (psi)	q_o (stb/day)
0	850
565	828
1130	788
1695	729
2260	651
2826	555
3000	522
5615	0



Construction of IPR curves using Test points

It has been shown that well IPR curves can be constructed using reservoir parameters including formation permeability, fluid viscosity, drainage area, wellbore radius and well skin factor. These parameters determine the constant (e.g. the P.I.) in the IPR model. However the values of these parameters are not always available. Thus the test points (measured values of production rates and flowing bottom hole pressure) are frequently used for constructing the IPR curves.

Constructing IPR curves using test points involves backing calculation of the constants in the IPR models. For a single phase (unsaturated oil) reservoir, the model constant, J^* can be determined by:

$$J^* = \frac{q_i}{\left(\bar{P} - P_{wfi} \right)}$$

Where q_i is the tested production rate at tested flowing bottom hole pressure P_{wfi}

For a partial two phase reservoir, model constant J^* in the generalized Vogel's equation must be determined based on range of tested flowing bottom hole pressure. If the tested flowing bottom hole pressure is greater than the bubble point pressure, the model constant J^* should be determined by :

$$J^* = \frac{q_i}{\left(\bar{P} - P_{wfi} \right)}$$

If the tested flowing bottom hole pressure is less than bubble point pressure the model constant J^* should be determined using

$$J^* = \frac{q_t}{\left(\bar{P} - P_b \right) + \frac{J^* P_b}{1.8} \left[1 - 0.2 \left(\frac{P_{wf}}{\bar{P}} \right) - 0.8 \left(\frac{P_{wf}}{\bar{P}} \right)^2 \right]}$$

Question 4

Construct IPR of the wells in an under-saturated oil reservoir using the generalized Vogel's equation, the following data is given:

Reservoir pressure $\bar{P} = 5000\text{psia}$

Bubble point pressure $P_b = 3000\text{psia}$

Tested flowing bottom hole pressure in well A $P_{wf1} = 4000\text{psia}$

Tested production rate from well A $q_1 = 300\text{ stb/day}$

Tested flowing bottom hole pressure in well B $P_{wf2} = 2000\text{psia}$

Tested production rate from well B $q_2 = 900\text{ stb/day}$

SOLUTION

Well A

Well A bottom hole flowing pressure is above the bubble point pressure

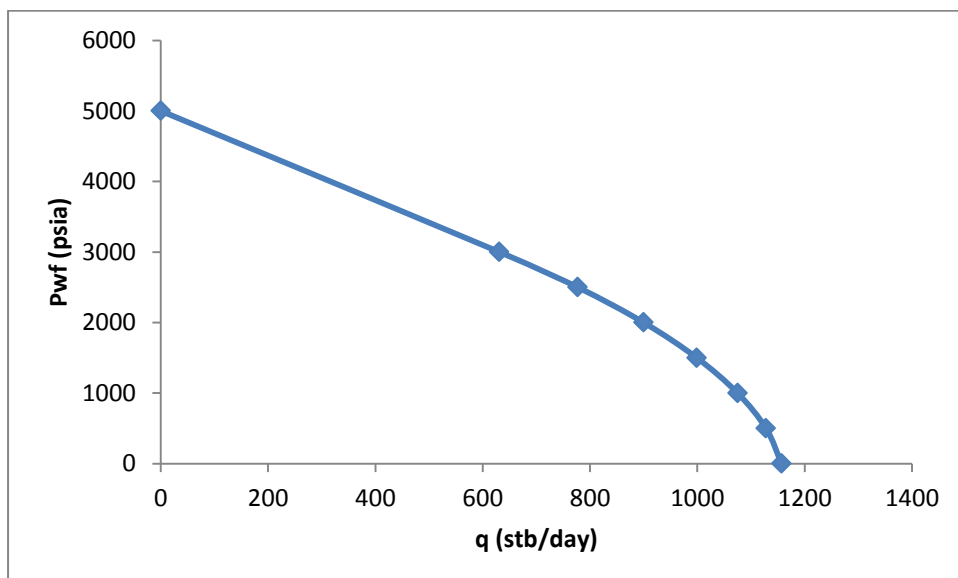
$$J^* = \frac{q_i}{\left(\bar{P} - P_{wfi} \right)}$$

$$= \frac{300}{5000 - 4000}$$

$$= 0.3000 \text{ stb/day-psi}$$

Calculated points are

P_{wf} (psia)	q (stb/day)
0	1100
500	1072
1000	1022
1500	950
2000	856
2500	739
3000	600
5000	0



Well B

Well B bottom hole flowing pressure is below the bubble point pressure

$$J^* = \frac{q_t}{\left(\bar{P} - P_b \right) + \frac{J^* P_b}{1.8} \left[1 - 0.2 \left(\frac{P_{wf}}{\bar{P}} \right) - 0.8 \left(\frac{P_{wf}}{\bar{P}} \right)^2 \right]}$$
$$\frac{900}{\left[(5000 - 3000) + \frac{3000}{1.8} \left(1 - 0.2 \left(\frac{2000}{3000} \right) - 0.8 \left(\frac{2000}{3000} \right)^2 \right) \right]}$$

$$= 0.3156 \text{ stb/day-psi}$$

Calculated points are

P_{wf} (psia)	q (stb/day)
0	1157
500	1128
1000	1075
1500	999
2000	900
2500	777
3000	631
5000	0

